



School of Information Technology

IT368C Robotic Process Automation

Assignment Guidebook AY2025 S2

1. Introduction

1.1 Learning Outcome

Upon successful completion of this project, students will be able to:

1. Explain what Robotics Process Automation and its challenges is
2. Use Robotics Process Automation tools to automate business processes
3. Test and debug automated processes

2. Assessment Scenario

2.1 Background

Validity Vision is a digital platform that enables clients to subscribe to specialist medical care through a customized online portal, offering convenient access to health and wellness services.

As a business analyst, you will support Validity Vision's automation goals using RPA. By creating a Process Design Document and a UiPath Attended Bot, the marketing team can streamline their manual tasks, including:

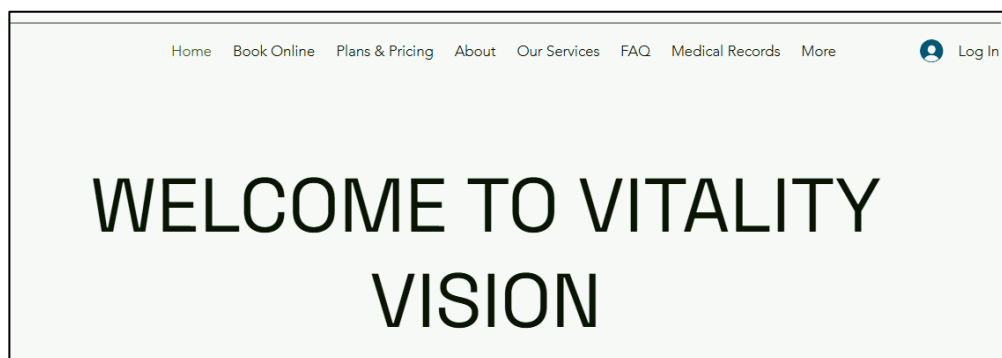
- Extracting daily updates on featured healthcare specialists and emailing the outreach division
- Retrieving records from the Medical Records Portal
- Entering client and care-related data into Validity Vision's Basic Medical Screening Data Bank via a web form
- Sending health-education and service materials based on client records
- Validating key data fields such as emails and clinical information

2.2 Deliverables

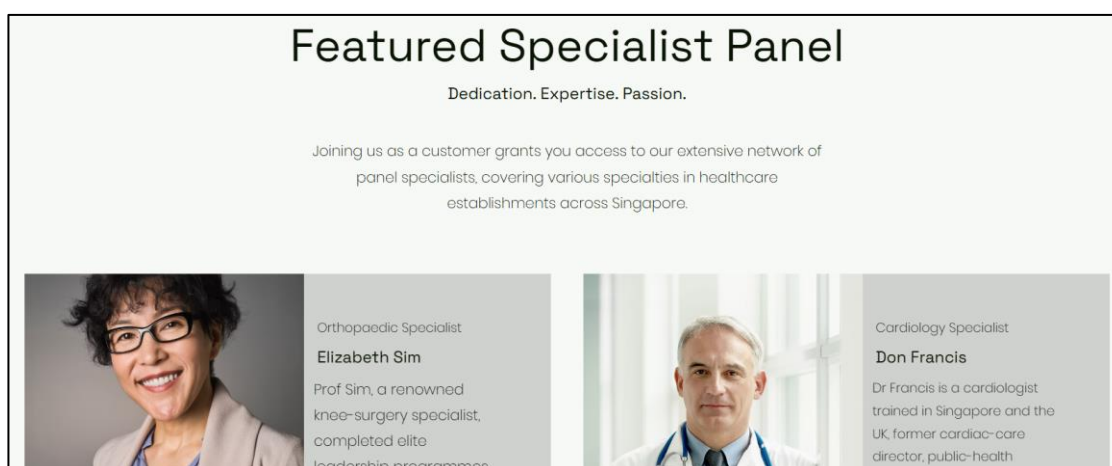
- **Process Design Document (PDD) (15 marks)** – template is available in Politemall, please submit in Word file format.
- **Attended Bot (60 marks)** – To automate the marketing team processes
 - Create UiPath project folder with all source code working on Edge browser
 - Export the generated emails from your Outlook and add them to your project folder. You may follow instructions from this link: [Save a message as a file - Outlook \(microsoft.com\)](#)
 - Zip your project folder with the emails and excel output file in a zip file for submission. Your zip file name should have the following format: adminNum_name.zip (Example: 033193M_Loo Kit Fan.zip)
- **Final Presentation (5 marks)** – Present the Project and Process Design Document in the final presentation.

2.3 Current Process Flow for extracting featured specialists from Featured Specialist Panel section:

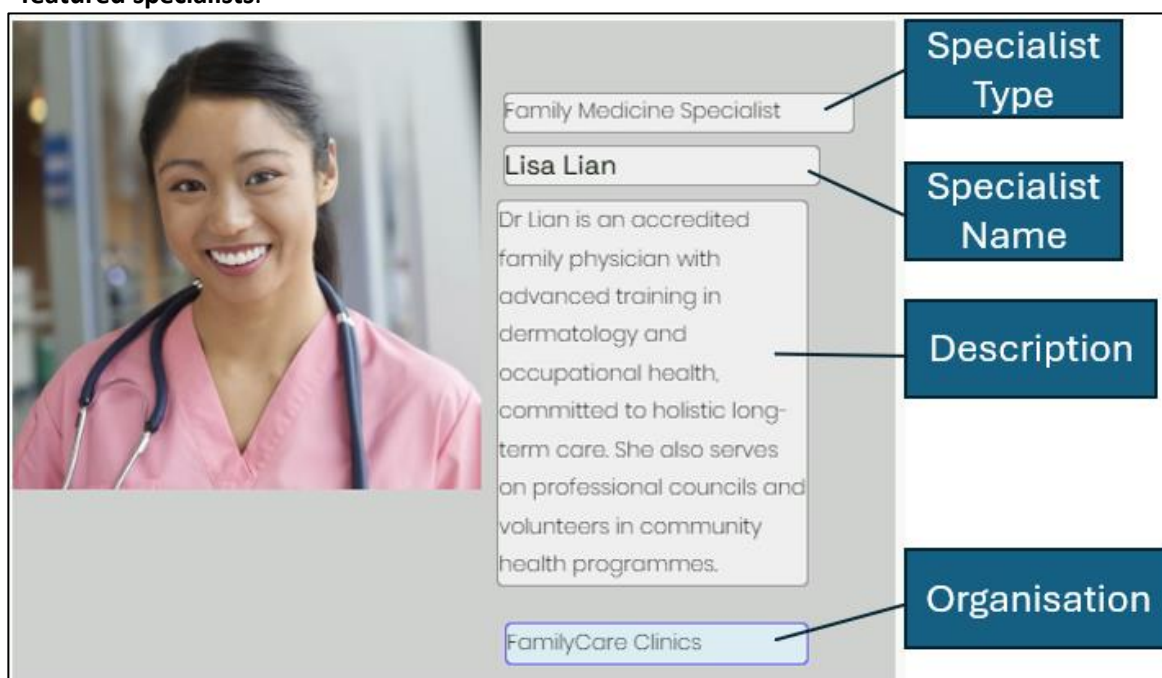
- 1) Marketing staff will access the **Vitality Vision website**: <https://lookitfan.wixsite.com/vitalityvision>



- 2) Staff will access the list of featured specialists by scrolling down to **Featured Specialist Panel** section.



- 3) Staff will then copy the **Specialist Type**, **Specialist Name**, **Description** and **Organisation** of the featured specialists.



- 4) Staff will then paste the information into an excel sheet and save it as “Featured Specialist (**today’s date**).xlsx”. Today’s date refers to the **date of the day the staff extracts the featured specialists’ information**. The excel sheet should contain column headers accordingly.

A	B	C	D
Specialist Name	Specialist Type	Description	Organisation
Elizabeth Sim	Orthopaedic Specialist	Prof Sim, a renowned	Elizabeth Hospital, Orthopaedic Department
Don Francis	Cardiology Specialist	Dr Francis is a cardiolo	TTSH, Department of Cardiology
Alexa Young	Healthcare Technologist	Alexa Low is a physicia	WitzHealth Tech Pte Ltd
Philip Eng	Thoracic Oncologist	Dr Eng is a senior respi	CGH, Oncology Department
Kevin Loo	Geriatric-medicine Specialist	Dr Loo is a geriatric-me	Beacon Health Group
Lisa Lian	Family Medicine Specialist	Dr Lian is an accreditec	FamilyCare Clinics

- 5) Staff will then email the “Featured Specialist (**today’s date**).xlsx” excel file to the publicity team. The email address is sitrapadec2019@gmail.com. The email content is as follows:



Send

To

Cc

☐ [SIT RPA <sitrapadec2019@gmail.com>](mailto:sitrapadec2019@gmail.com)

Subject

<<Vitality Vision> Featured Specialists



Featured Specialist (11-30-2025).xlsx
 20 KB

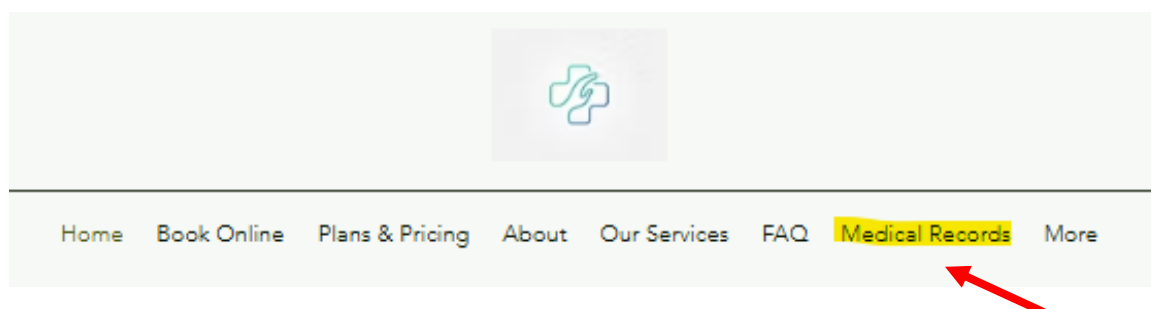
Dear publicity team,

Please see attached file for the featured Specialists for 11 Nov 2025.

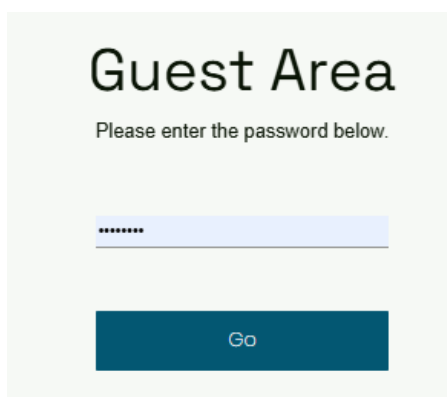
Thank you.

2.4 Current Process Flow for extracting customer records from Medical Records Portal and entering into webform:

- 1) Staff will **access** the staff access portal by clicking on **Staff Access Portal** in the top menu:



- 2) Staff will then enter the password (**staff123**) for the access to the **Staff Access Portal**:



Guest Area

Please enter the password below.

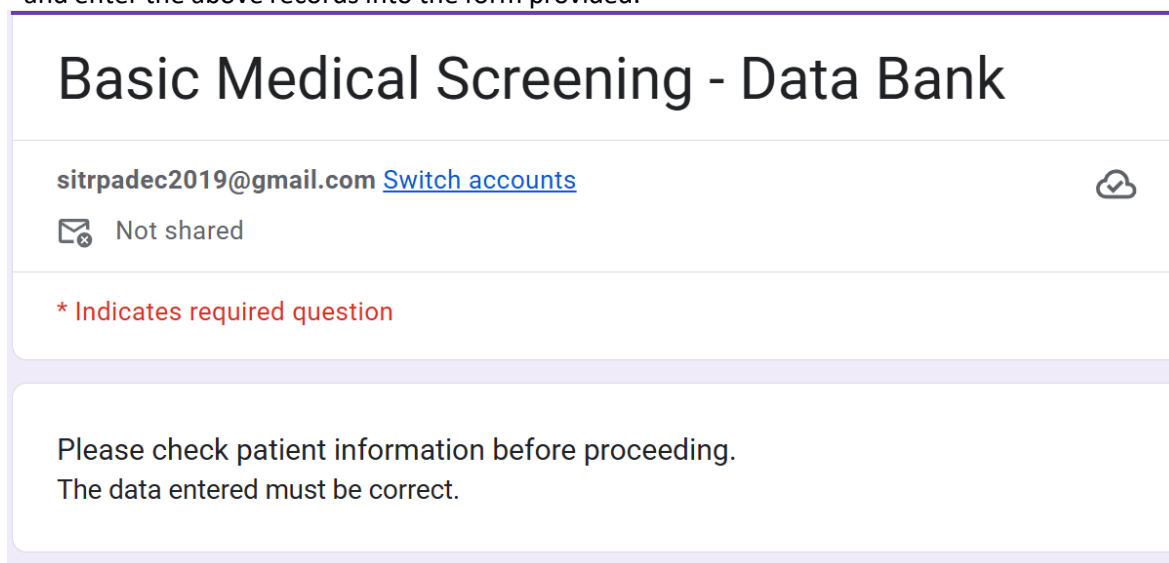
Go

- 3) Staff will then **copy out the data** from the following table and **save the data** into an **excel sheet**:

Name	Age	Gender	On Medication	Medication	BP Family History	Smoke	Drink	Height (m)	Weight (Kg)	Email Address	Attending Clinic	Clinic Email Address	Last date of visit
Peter Tan Min Shun	26	Male	Yes	Lovastatin	Yes	No	Yes	1.67	80	pete@gmail.com	Ang Mo Kio Vitals	Vital@vitalis.co	7/10/2025
Jane Chee Mei Lin	48	Female	No		Yes	Yes	Yes	1.7	57	ejane@hotmail.c	General Vitals (Bishan)	VBishan@vitalis.co	8/9/2025
Jumail bin Abdul Ghani	48	Male	Yes	Lovastatin	Yes	No	No	1.72	60	rp@gmail.com	Ang Mo Kio Vitals	Vital@vitalis.co	9/11/2025
Fahd bin Abdul Quddoos	40	Male	Yes	Allopurinol	Yes	Yes	No	1.65	62	ahd@hotmail.co	Singapore Opto Specialist	SGopto@gmail.com	10/11/2025
Mahmood bin Maazin	35	Male	No		No	Yes	No	1.6	77	ahmood@gmail.c	General Vitals (Bishan)	VBishan@vitalis.co	11/11/2025

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- 4) Staff will then **access** the **Basic Medical Screening - Data Bank** (<https://forms.gle/waborobYGFHcov9aA>) and enter the above records into the form provided:



Basic Medical Screening - Data Bank

sitrapadec2019@gmail.com [Switch accounts](#)

 Not shared

*** Indicates required question**

Please check patient information before proceeding.
The data entered must be correct.

- 5) Staff will also check and ensure the following fields meet the rules criteria **BEFORE** entering the data into the web form. If there is any wrong data, the staff will not enter the data into the form.

The staff will then move to next record until all valid records are entered into the form.

Rules	Example
Ensure that Age is a number that can be valid as a person's age	Valid: 18, 25, 72 Invalid due to not being numeric: Twenty, fifty Invalid due to numbers being out of range: 1000, 0
Ensure that Height is within reasonable range	Valid: 1.54, 1.82, 1.72 Invalid due to not being numeric: Twenty, fifty Invalid due to numbers being out of range: 172, 182 (This is typically due to entering the height in CM unit instead of Metre unit)
Ensure that email address is in valid format	Valid formats: testingemail@gmmail.com, test123@yahhhooo.com Invalid formats: testingEmailGmmail.com, test123@yahoooohh Please ignore wrong domains. Only validate the format of the email address.
Any other validations that may be applied to other fields. You may implement more validations.	Example: "Gender" to have only Male and Female "Weight" to be within reasonable range "Drink" and other fields to contain correct options example: Yes, No, Unsure And other validations for other fields.

2.5 Current Process Flow for sending promotional emails:

- 1) Using the same **medical records**, send **promotional emails** to **customers based on the conditions below**. The rules are as follows:

Condition	Email sample
If Customer's BMI is above 25 and BP Family History is Yes	Subject Discover Better Health: Seminar Invite  Vitality Essentials: Your Guide to Better Health: 311 KB Send as Adobe Document Cloud link Yes No ? Dear Peter Tan Min Shun, We're excited to share an exclusive preview of our new medical health booklet, "Vitality Essentials - Managing Blood Pressure & Weight for Better Health" — attached as a PDF. As a valued customer, we'd also like to invite you to our upcoming online seminar on blood pressure and weight management, where you'll gain practical strategies for long-term wellness. Enjoy a special 10% discount when you sign up. Reply to this email to reserve your spot and claim your discount. Thank you for being a part of our community! Regards, • Customer's name should be in the email. • You may modify the email with photos/ company logo if you wish to. • Email Attachment can be downloaded from Politemall.
If customer's attending clinic matches any of organisation in the featured specialist list	Subject Your Attending Specialist Takes the Spotlight! Hi [Customer's Name], We're excited to share today's Featured Specialist Panel—highlighting the clinician(s) who will be attending to your care. <i>[May insert an image / name of the featured clinician whose organisation matches the attending clinic of customer]</i> Your featured specialist brings trusted expertise and evidence-based insight to help you stay on top of your health. You can also look forward to exclusive health tips, wellness guidance, and updates on upcoming seminars designed to support better long-term health. Stay tuned for more valuable resources coming your way soon! Regards, • Customer's name should be in the email. • You may include details of the attending clinic/organisation and specialist name. • You may modify the email with photos/ company logo if you wish to.

Other details: Details for PDD

The staff typically starts the process in the morning at 9am daily during working days (Monday to Friday). The staff takes 7 minutes to complete entering each record into the webform.

There is usually an increase of transaction records during the peak period at end of the year. It is expected that there will be 90-100 records per day this year, and an expected increase of 10% in the following year. The marketing team have 3 Full Time Employees (FTEs) supporting this function.

Additional Features (where applicable)

- Features to enhance security of the application, such as using prompts or Orchestrator for password
- Emailing of status, logs, exceptions to administrator email after process is completed
- Using a configuration excel file to read fields such as URLs, administrator email address
- Adding Company Logo and other relevant information (contacts etc) to emails
- Other exceptions rules added for more data validation, such as checking for empty fields

3. Assessment

The assessment comprises of a total base mark of 80 marks and consist of **40%** of the total module mark.

3.1 Assessment Components

Deliverables		Total
Process Design Document • Introduction and purpose • As-Is Process and screenshots • To-Be Process and exceptions	3 marks 5 marks 7 marks	15 marks
Project and PDD Presentation • Presentation <i>Presentation will be one to one with tutor, 10 minutes total.</i>	5 marks	5 marks
Attended Bot in UiPath • Extracting featured specialists from Featured Specialist Panel section • Extracting records from medical portal and entering into webform • Emailing promotional materials according to record details • Project organization, such as naming, annotations and flowcharts	15 marks 20 marks 20 marks 5 marks	60 marks
Total		80 marks

3.2 Assessment Rubrics**Categories:**

- 1) Poor and Developing: **Below 50%** of marks allocated to category
- 2) Good and Excellent: **Above 50%** of marks allocated to category

Process Design Document (15 Marks)				
Category	Poor	Developing	Good	Excellent
Introduction And purpose (3 Marks)	Provide little analysis, description, and relevant key facts of the objectives of the process (0 – 0.5 marks)	Provide 1-2 detailed analysis, description and relevant key facts of the objectives of the process (1 – 1.5 marks)	Provide 3-4 detailed analysis, description and relevant key facts of the objectives of the process (2 – 2.5 marks)	At least 5 detailed analysis, description and relevant key facts of the objectives of the process (3 marks)
As-Is Process And screenshots (5 Marks)	Provide little documentation in the As-Is process without screenshots and no detailed Process Map (0 - 1 marks)	Provide some documentation in the As-Is process with screenshots and detailed Process Map (1.5 – 2.5 marks)	Provide detailed documentation in the As-Is process with screenshots and detailed Process Map (3 – 4 marks)	Provide detailed documentation in the As-Is process with screenshots and detailed Process Map. Provide clear understanding in the detailed Process Map (4.5 - 5 marks)
To-Be Process and exceptions (7 Marks)	Provide little documentation in the To-Be Process Map without business exceptions (0 – 1.5 marks)	Provide some documentation in the To-be Process Map with 1-2 business exceptions (2 – 3.5 marks)	Provide detailed documentation in the To-Be Process Map with 3-4 business or more business exceptions (4 – 5.5 marks)	Provide detailed documentation in the To-Be Process Map with 3-4 or more business exceptions as well as handling of system exceptions (6 - 7 marks)
Attended Bot in UiPath (60 marks)				
Category	Poor	Developing	Good	Excellent
Extracting featured specialist from Featured Specialist section (15 marks)	Implement with errors. Wrong/no extraction of data from site. Error in sending email to publicity team. (0 – 3.5 marks)	Implement with no errors. Correct extraction of data from site but creation of excel records with wrong headers Some wrong information such as wrong date in email sent to publicity team. (4 – 7.5 marks)	Implement with no errors. Correct extraction of data from site and creation of excel records with correct headers All correct information in email to publicity team. (8 – 11.5 marks)	Implement with no errors. Correct extraction of data and creation of excel records with correct headers. All correct information in email to publicity team. Using configuration file to read fields such as URLs, Publicity team email address. (12 - 15 marks)

Extracting records from medical portal and entering into webform (20 marks)	Implement with errors. Wrong/no extraction of data from site. Little justification in supporting the functionality of the application, and unable to submit the customer records correctly. (0 – 5 marks)	Implement with no errors. Hardcoded password for staff login. Correct extraction of data from site Able to submit customer's records with 3 or less validation checks Process stops when data is incorrect, instead of moving to next record. (5.5 – 10 marks)	Implement with no errors. Features to enhance security of the application, such as using prompts for password. Correct extraction of data from site Able to submit customer's records with 4 - 5 validation checks Process continues to next record when data is incorrect Emailing of simple status to administrator email after process is completed (10.5 - 15 marks)	Implement with no errors. Features to enhance security of the application, such as using prompts or Orchestrator for password. Correct extraction of data from site Able to submit customer's records with 6 or more validation checks Emailing of detailed status, logs, exceptions to administrator email after process is completed, with details of exceptions in a comprehensive email (15.5 - 20 marks)
Emailing promotional materials according to record details (20 marks)	Implement with errors. Unable to send emails. Have little to some email templates, with wrong information Email sent but to wrong customers (0 – 5 marks)	Implement with no errors. Able to send email to correct customers but email rules contain 2 or more wrong information (5.5 – 10 marks)	Implement with no errors. Able to send email to correct customers with only 1 or no wrong information. (10.5 - 15 marks)	Implement with no errors. Able to send email to correct customers with no wrong information. Additional photos of logo or customer's photo in email (15.5 - 20 marks)
Project organization, such as naming, annotations, and flowcharts (5 marks)	Poor organization of project such as bad naming of variables, files, and sequences with little to no annotation (0 – 1 mark)	Standard organization of project. Proper naming of variables, files, and sequences with some annotations. Uses flowchart. (1.5 – 2.5 marks)	Good organization of project with proper naming of variables, files and sequences. Proper annotations at higher level sequences. Using flowchart to organize project and workflow (3 – 3.5 marks)	Good organization of project with proper naming of variables, files and sequences. Proper annotations at higher level sequences. Using flowchart to organize project with invoking of files with proper arguments. (4 - 5 marks)

Presentation (5 Marks)				
Category	Poor	Developing	Good	Excellent
Project presentation (5 Marks)	Unable to clearly explain any of the As-Is, To-Be workflows and lack clarity in flow of the presentation of project (0-1 mark)	Able to present and explain the As-Is, To-Be workflows and exceptions, but too technical for non- technical staff to understand (marketing team) (1.5 – 2.5 marks)	Able to clearly explain the As-Is, To-Be workflows and exceptions clearly with good examples and clarity (3 – 3.5 marks)	Able to clearly explain the As-Is, To-Be workflows and exceptions clearly with good examples and clarity, and able to present confidently and concisely such that the target audience can clearly understand the presentation (4 - 5 marks)